

Statistical Case Studies - Predicting Hotel Booking Cancellations

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1 Introduction

The hotel industry heavily relies on accurate demand forecasting and revenue management strategies to optimize occupancy rates and maximize revenue. However, hotel booking cancellations pose a significant challenge to this process. Cancellations lead to lost revenue, reduced occupancy rates, and increased operational costs. Therefore, predicting and understanding the factors that influence booking cancellations is crucial for hotels to make informed decisions and implement effective strategies to mitigate their impact. This study aims to develop and compare different statistical models for predicting hotel booking cancellations. The findings of this project contribute to the understanding of factors influencing hotel booking cancellations and provide valuable insights for hotel revenue management. These insights can assist hotel managers in making informed decisions, such as overbooking strategies, pricing adjustments, and targeted marketing campaigns, to minimize the negative impact of cancellations on hotel revenue and operations.

2 Exploratory Data Analysis

The original dataset contains 119210 entries. Our exploration of the data began with data cleaning, ensuring the integrity and usability of the dataset.

N/A value replacement: 4 missing entries were noted in the **children** variable. We addressed these by applying mode imputation to preserve the most common values in the dataset, thereby ensuring consistency.

Outlier Reduction: We treated entries lying outside the 1st and 99th percentiles of each numerical attribute as outliers. 6734 entries were excluded after considering the outliers.

2.1 Descriptive Statistics

Following the data cleaning process, an analysis of the refined dataset, which includes 112,476 entries, provides detailed descriptive statistics for various booking parameters. The lead time, measuring the interval from booking to arrival, averages 100.58 days, with a standard deviation of 99.58, indicating considerable variability in booking behaviors. Weekend and weekday stays average 0.89 and 2.39 nights respectively, suggesting short-term bookings are common. The dataset predominantly features bookings by adults, averaging 1.86 per reservation, with minimal child (average 0.09) or baby guest occurrences, aligning with typical hotel booking demographics. A notably low mean of 0.05 for previous cancellations highlights the rarity of repeat cancellations among guests, while the mean for previous bookings not canceled is also low at 0.03, suggesting a significant portion of first-time or infrequent guests. Requests for parking spaces and special requests occur in 5.93% and 55% of bookings, respectively, indicating potential areas for enhancing guest satisfaction and loyalty. The average daily rate

is approximately \$100.69, with a significant range from the minimum to the maximum, reflecting a diverse pricing strategy.

	Count	Mean	Std	Min	25%	50%	75%	Max
lead_time	112476.0	100.58	99.58	0.0	18.0	69.0	157.0	444.0
stays_in_weekend_nights	112476.0	0.89	0.9	0.0	0.0	1.0	2.0	4.0
stays_in_week_nights	112476.0	2.39	1.56	0.0	1.0	2.0	3.0	9.0
adults	112476.0	1.86	0.47	1.0	2.0	2.0	2.0	3.0
children	112476.0	0.09	0.37	0.0	0.0	0.0	0.0	2.0
babies	112476.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
previous_cancellations	112476.0	0.05	0.22	0.0	0.0	0.0	0.0	1.0
previous_bookings_not_canceled	112476.0	0.03	0.24	0.0	0.0	0.0	0.0	3.0
required_car_parking_spaces	112476.0	0.06	0.24	0.0	0.0	0.0	0.0	1.0
total_of_special_requests	112476.0	0.55	0.76	0.0	0.0	0.0	1.0	3.0
avg_daily_rate	112476.0	100.69	44.17	0.0	70.0	95.0	125.1	250.0

Figure 1: Data description after data cleaning

2.2 Bivariate Analysis

2.2.1 Correlation for predictor variables

An examination of the intercorrelations between predictor variables is an essential step to ensure the robustness of the statistical models we aim to construct. To facilitate this analysis, a heatmap Figure 2 has been generated, presenting the correlation coefficients (ρ) among predictors that exhibit a significant degree of association, defined by $|\rho| \geq 0.5$. This visual tool allows us to identify collinear variables that could potentially distort the model's performance.

In Figure 2, the heatmap indicates a perfect correlation ($\rho = 1$) between **arrival_date_week_number** and **arrival_date_month**, which is expected due to their shared role in representing time-related aspects of the booking date. Given their redundancy, we opt to exclude **arrival_date_month** from our modeling efforts.

Conversely, certain pairs of variables such as **hotel_City** and **hotel_Resort**, as well as **deposit_type_No_Deposit** and **deposit_type_Non_Refund**, display a high negative correlation, indicating a clear binary relationship, where each variable is the opposite category of the other. These variables are exclusive options within their categories, together representing all possible scenarios. For this reason, they are important for our analysis and we will keep them. This approach ensures we capture the full picture of the dataset's categories and understand how they relate to cancellation trends.

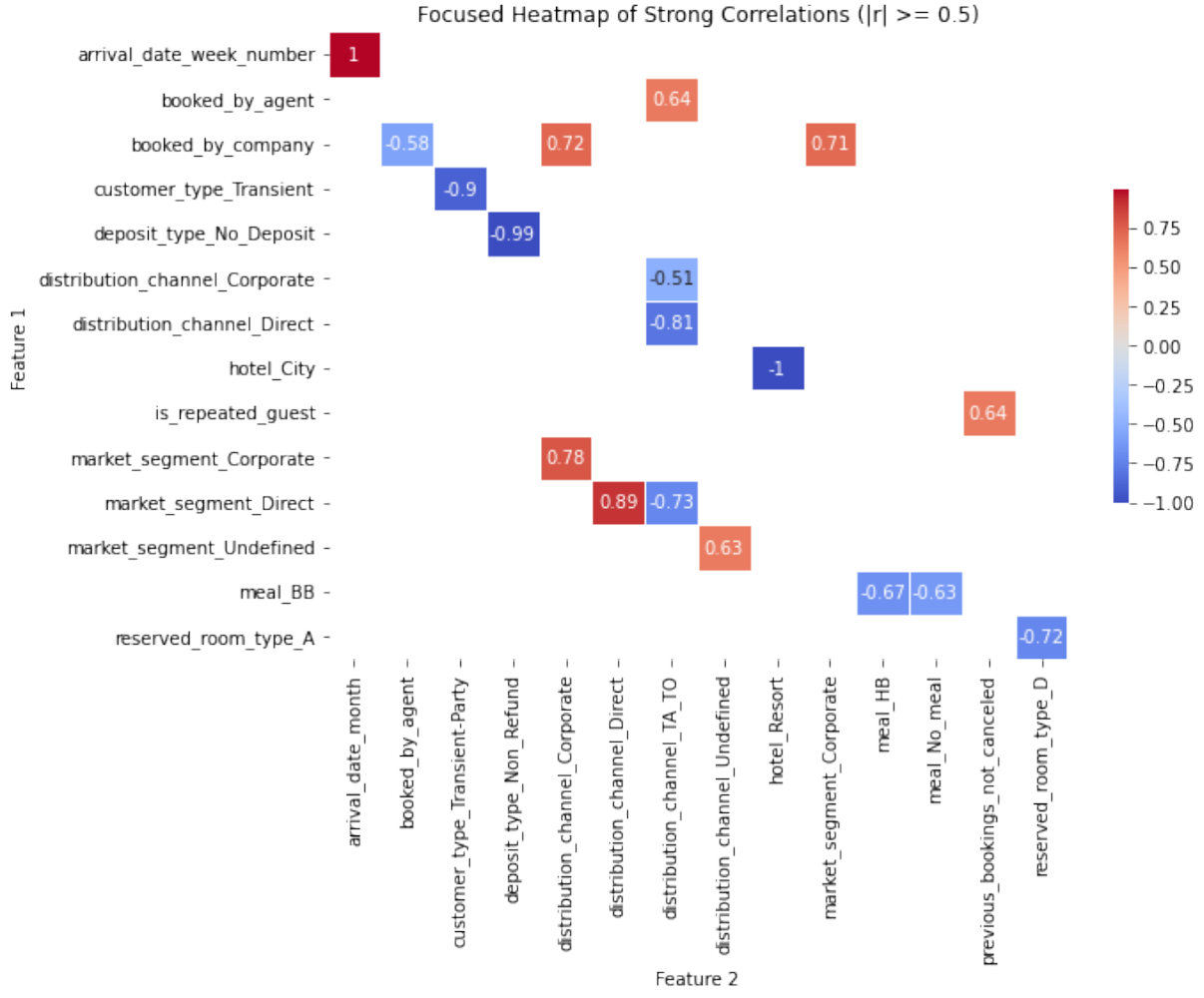


Figure 2: Correlation Heatmap for predictor variables with high correlation

2.2.2 Correlation with Target Variable

In our bivariate analysis of the hotel bookings dataset, we examine the relationships between each predictor variable and the target variable, **is_canceled**. We calculated Pearson correlation coefficients for continuous predictors to identify linear associations and used chi-square tests for categorical predictors to evaluate the strength and significance of their associations with cancellation outcomes. This analysis strategically focuses on identifying variables that show significant correlations with the likelihood of cancellation, considering only correlation coefficients greater than 0.1 or less than -0.1 as relevant. Additionally, we provide a visual representation in Figure 3 using a heatmap to display these coefficients, which maps out the strength and direction of relationships between various predictors and the target variable, offering a clear and immediate visual interpretation of which factors are most strongly associated with cancellations.

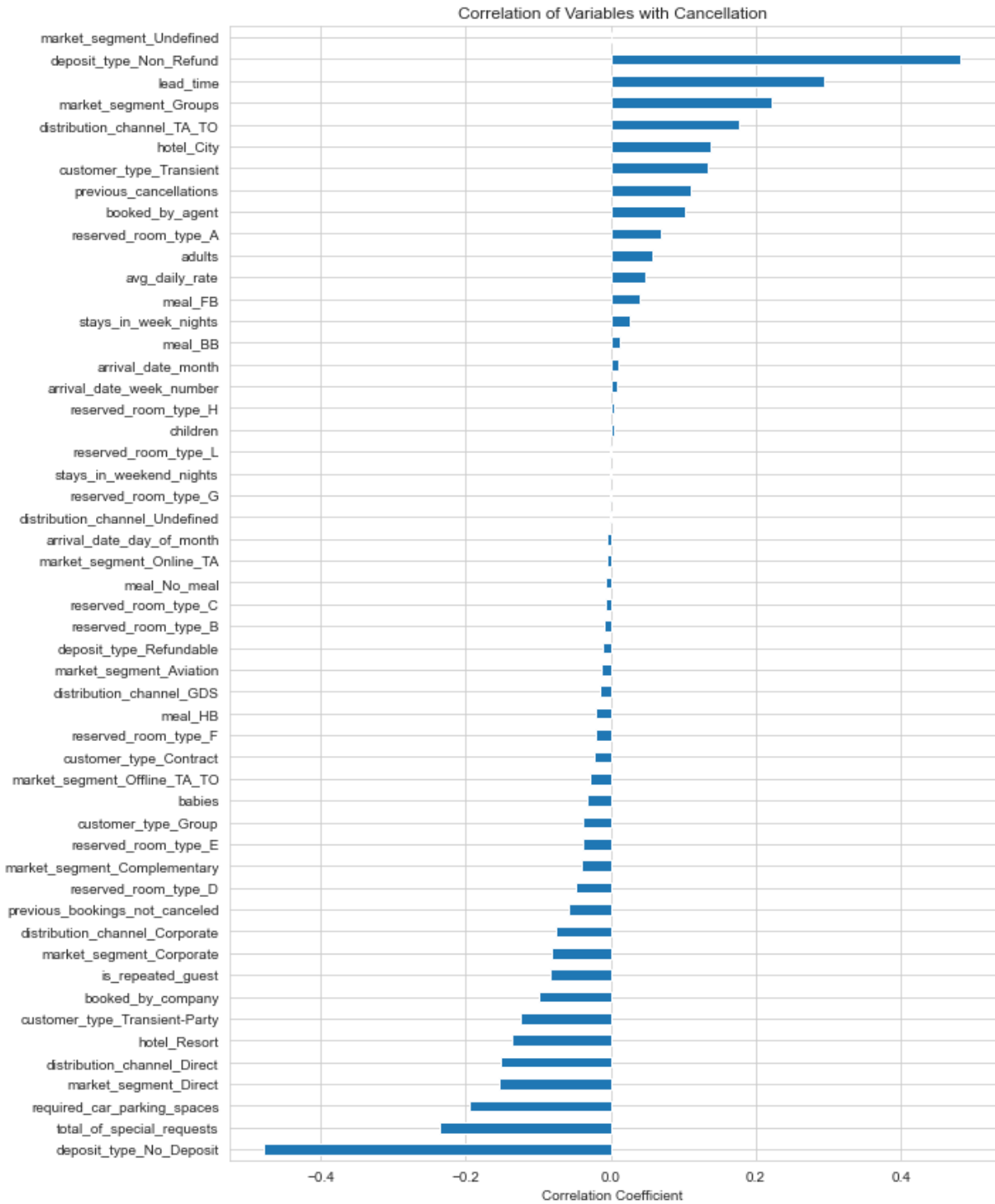


Figure 3: Correlation of Variables with Cancellation

2.2.3 Selection of Features for In-depth Analysis

After a detailed evaluation of the correlations between predictors and the **is_canceled** target variable, we have carefully organized the variables into distinct categories. This organization streamlines our analysis and focuses on key areas likely to provide insights into the dynamics of booking cancellations. The table below clearly outlines these categories with their respective features, providing a structured framework for further analysis:

Table 1: Feature Categories and Details

Feature Category	Features
Target Variable	is canceled
Booking Details	lead time, arrival date week number, arrival date day of month, arrival date month, stays in weekend nights, stays in week nights, adults, children, babies, required car parking spaces, total of special requests, avg daily rate
Customer Information	is repeated guest, previous cancellations, previous bookings not canceled
Booking Source	booked by company, booked by agent, market segment X, distribution channel X
Hotel Information	hotel City, hotel Resort
Meal Information	meal BB, meal HB, meal FB, meal No Meal
Room Information	reserved room type X
Deposit Information	deposit type X
Customer Type	customer type Contract, customer type Group, customer type Transient, customer type Transient-Party

Following the analysis of the statistical relationships between predictors and the target variable **is canceled**, we have identified a set of features for comprehensive analysis. These features are selected based on their correlation with the cancellation probability, with correlation coefficients greater than 0.1 or less than -0.1, indicating a moderate to strong linear association. This selection process allows us to focus on the most influential factors affecting hotel booking cancellations, aiming to derive actionable insights to mitigate cancellation rates.

Booking Details

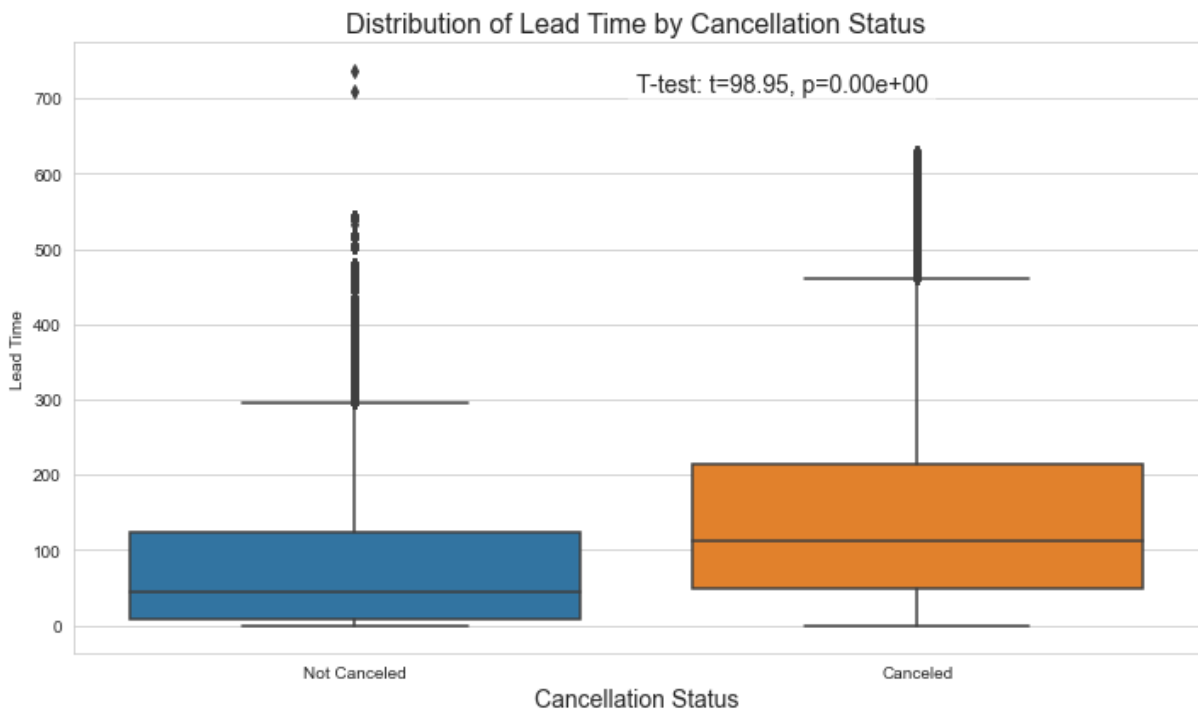


Figure 4: Leadtime

- **Lead Time (Correlation: 0.292930):** The number of days between the booking date and the arrival date.
- **Required Car Parking Spaces (Correlation: -0.195696):** Number of parking spaces that were requested by the customer.
- **Total of Special Requests (Correlation: -0.234925):** Number of special requests that were made by the customer.

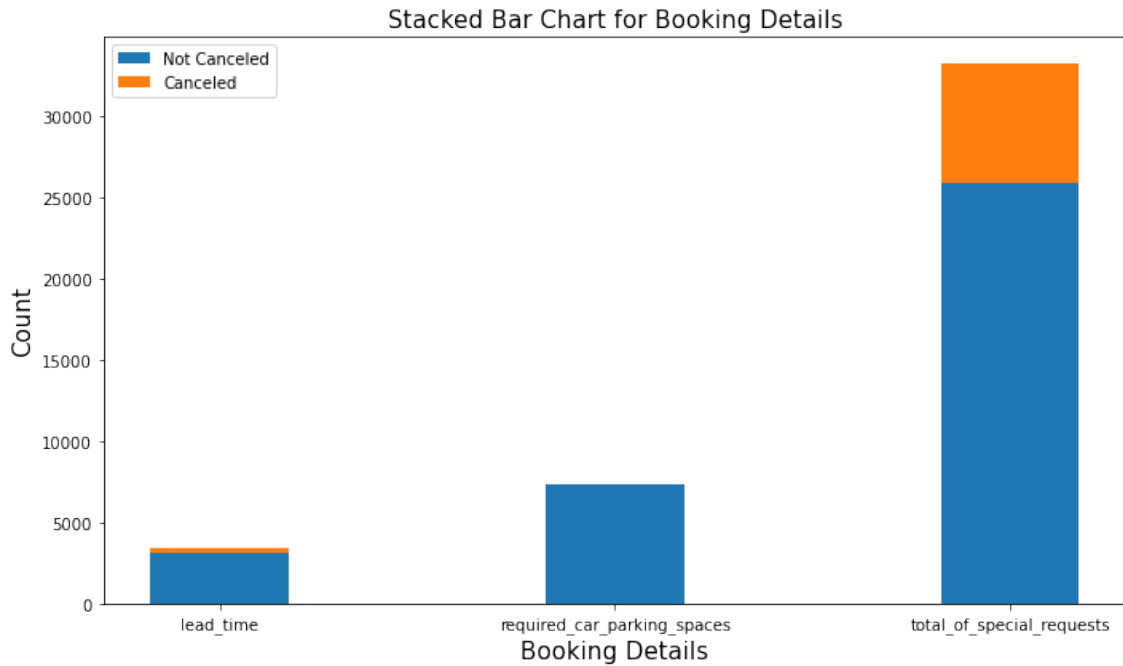


Figure 5: booking detail

Hotel Information

- **hotel_City (Correlation: 0.137056) and hotel_Resort (Correlation: -0.137056):** Indicates whether the booked hotel is a city hotel or a resort hotel.

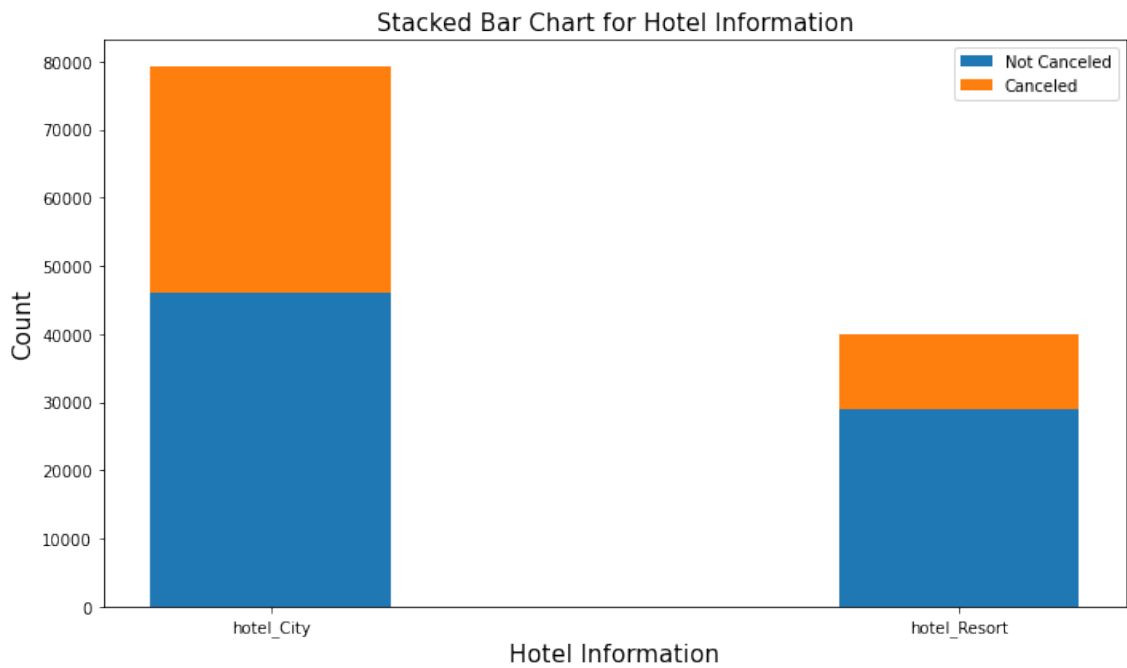


Figure 6: Hotel Information

Booking Source (Extended)

- **market_segment_Direct**(Correlation: **-0.154397**), **market_segment_Groups**(Correlation: **0.222016**), **distribution_channel_Direct**(Correlation: **-0.151571**) and **distribution_channel_TA_TO**(Correlation: **0.176117**): Several indicator variables indicate the market segment designation or the distribution channel.

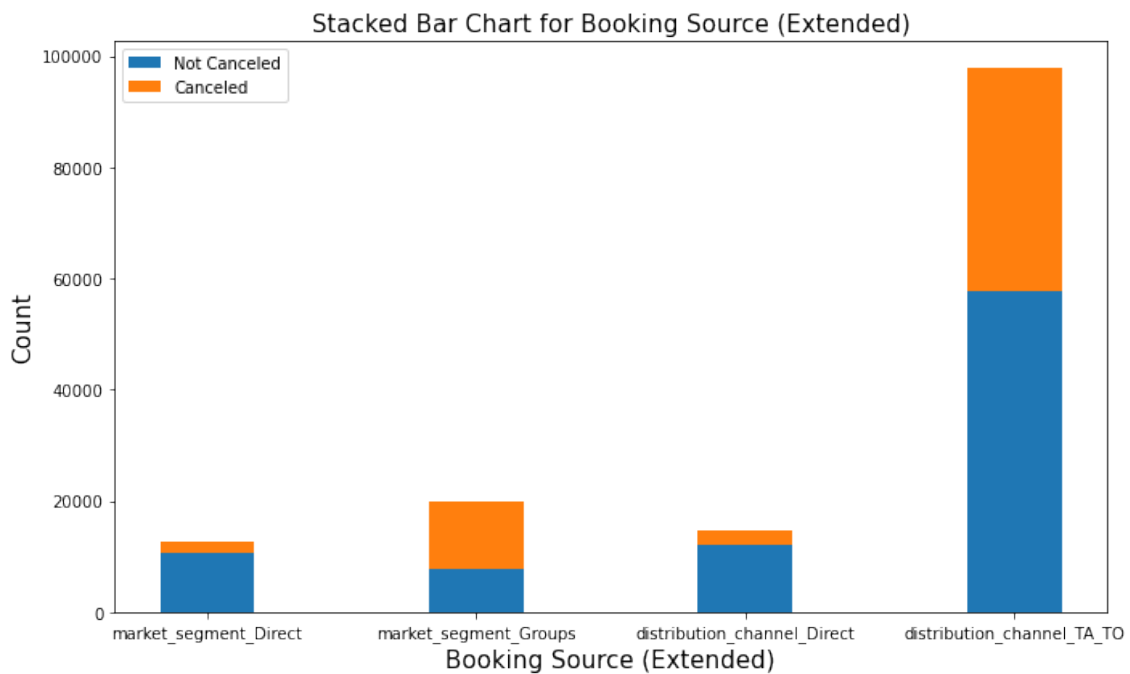


Figure 7: Booking source

Deposit Information

- **deposit_type_No_Deposit**(Correlation: **-0.477989**) and **deposit_type_Non_Refund**(Correlation: **0.481538**): Indicates information about whether a deposit was made.

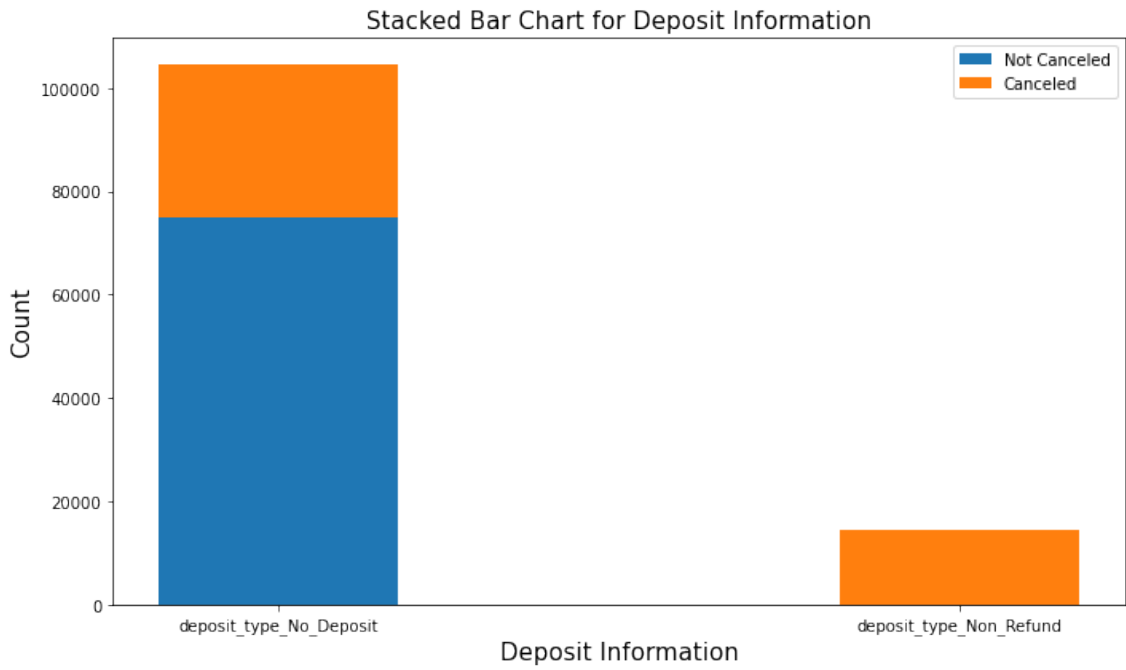


Figure 8: Deposit Information

Customer Type

- **customer_type_Transient**(Correlation: 0.133321) and **customer_type_Transient-Party**(Correlation: -0.124367):

Indicates if the booking is not part of a contract or group, and is not associated to other transient booking; Indicates if the booking is transient but is associated to another transient booking.

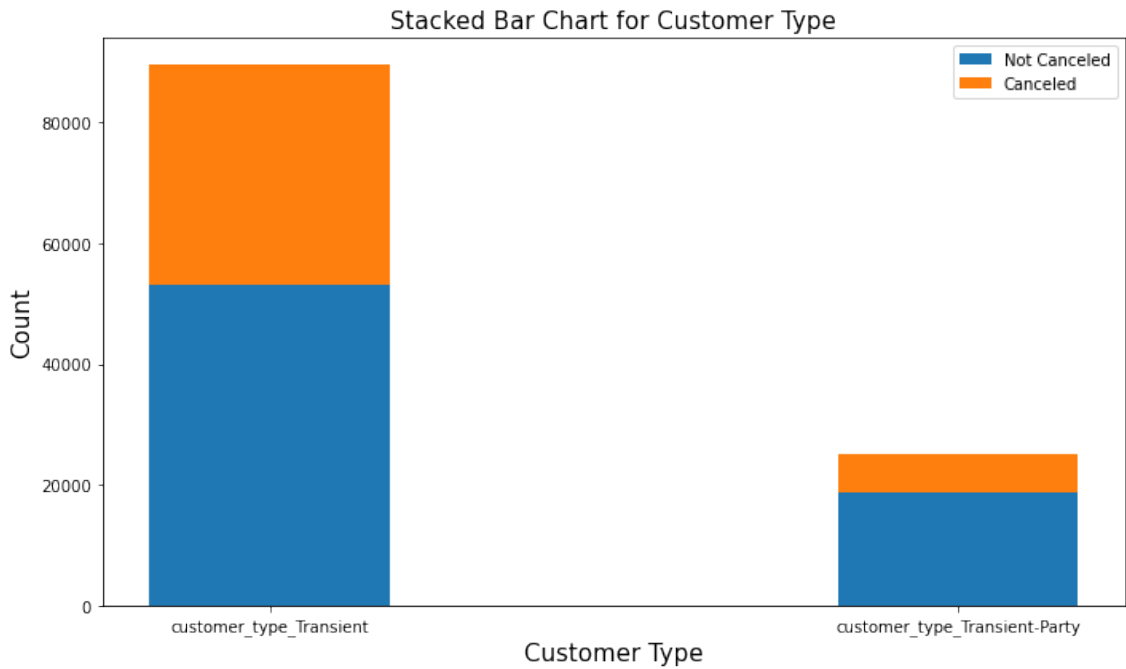


Figure 9: Customer Type

Our bivariate analysis, including visual data interpretation, identifies key predictors for the `is_canceled` variable. In Figure 4, **The Lead Time** shows a significant correlation of 0.292930, with box plots revealing a stark contrast between canceled and non-canceled bookings, statistically confirmed ($t=98.95$,

$p < 0.001$). This suggests that strategies like staggered non-refundable deposits or incentives for early bookings could reduce cancellations linked to longer lead times.

Figure 5 shows that **Required Car Parking Spaces** and **Total of Special Requests**, with negative correlations of -0.195696 and -0.234925 respectively, predict higher guest retention. This suggests that guests with car and specific requests are less likely to cancel, indicating that enhancing service personalization could strengthen guest commitment.

Figure 8’s stacked bar chart of **Booking Source** shows booking trends, highlighting that Travel Agents and Tour Operators (`distribution_channel_TA_TO`) lead in total bookings and cancellations. In contrast, the direct market segment (`market_segment_Direct`) and direct distribution channel (`distribution_channel_Direct`) exhibit lower cancellation rates. This suggests that direct customer interactions reduce cancellation risks, advising hotels to invest more in these channels than in intermediary agents.

The **Deposit Information** in Figure 9 highlights a paradox: higher cancellation rates for non-refund deposits (correlation: 0.481538) as shown in Figure 8, challenging traditional deterrent strategies. This calls for a reevaluation of deposit policies using a data-driven approach.

The **Customer Type analysis** (Figure 10) shows distinct cancellation rates between transient customers (correlation: 0.133321), typically individual bookings, and transient-Party customers (correlation: -0.124367), part of group bookings as seen in Figure 9. This suggests hoteliers need tailored strategies: personalized communication and flexible options may reduce cancellations among transient customers, while group incentives might better suit transient-Party customers.

3 Baseline Model and Logistic Regression

3.1 Baseline Model

A simple baseline model was created to predict every outcome as the majority class (not canceled). This model achieved an accuracy of 62.92%, setting a benchmark for more sophisticated models to surpass.

3.2 Logistic Regression

Logistic regression is a statistical model commonly used for binary classification tasks. It predicts the probability of an event occurring based on input features. In this case, we fit a basic logistic regression model to predict hotel booking cancellations. The performance metrics for the basic logistic regression model are presented in Table 2.

Table 2: Performance metrics for the Basic Logistic Regression

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.8035	0.8318	0.6128	0.7057

The basic logistic regression model achieved an accuracy of 80.35%, significantly outperforming the baseline model. The precision, which measures the proportion of true positive predictions among all positive predictions, was 83.18%. The recall, indicating the proportion of actual positive instances correctly identified, was 61.28%. The F1 score, a balanced measure of precision and recall, was 70.57%.

These results suggest that the logistic regression model provides a substantial improvement over the baseline, but there is still room for further enhancement in terms of recall and overall performance.

4 Interaction Terms and Non-linear Transformations

In this section, we introduce interaction terms and non-linear transformations to our logistic regression model. These enhancements aim to capture more complex relationships and improve model accuracy.

4.1 Interaction Terms

- **lead_time and market_segment combined to form**

lead_time_x_market_segment_Corporate: This interaction investigates the hypothesis that the influence of lead time on cancellation probability varies across different market segments. The rationale is that customers from different segments may exhibit diverse booking and cancellation behaviors.

- **total_of_special_requests and customer_type_Transient combined to form**

special_requests_x_customer_type_Transient: This term investigates whether transient customers with special requests are less likely to cancel, suggesting that such requests may indicate greater commitment to the stay.

- **previous_cancellations and booked_by_agent combined to form**

previous_cancellations_x_booked_by_agent: This interaction evaluates whether a history of cancellations combined with agent bookings affects cancellation probabilities differently, potentially revealing patterns specific to agent bookings.

4.2 Non-linear Transformations

In predictive modeling for booking cancellations, some variables showed non-linear relationships with the response variable based on their distributions. To better capture these dynamics and potentially improve model performance, non-linear transformations were applied.

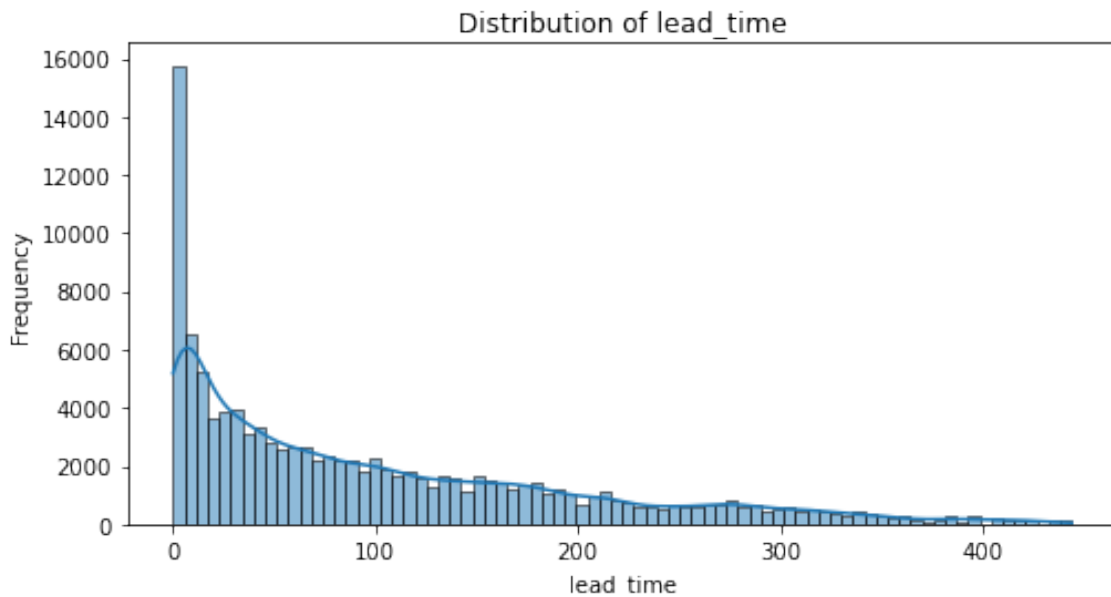


Figure 10: Distribution of lead_time

- **log_lead_time:** A logarithmic transformation was applied to the right-skewed lead time distribution to reduce the impact of outliers and more accurately model the decreasing influence on cancellation probability as lead time increases, supporting the hypothesis that cancellation likelihood increases at a diminishing rate over time.

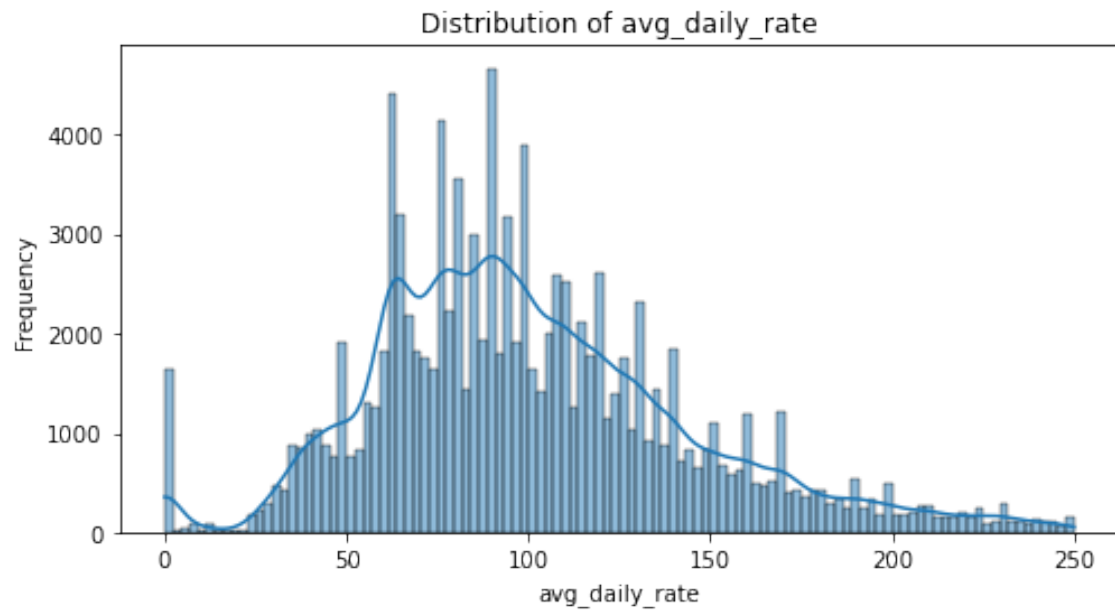


Figure 11: Distribution of `avg_daily_rate`

- **`avg_daily_rate_scaled`:** The average daily rate showed a right-skewed distribution, less pronounced than lead time, with large numerical values compared to other variables, potentially affecting model performance. To mitigate this, we divided the variable by 10 through a non-linear transformation.

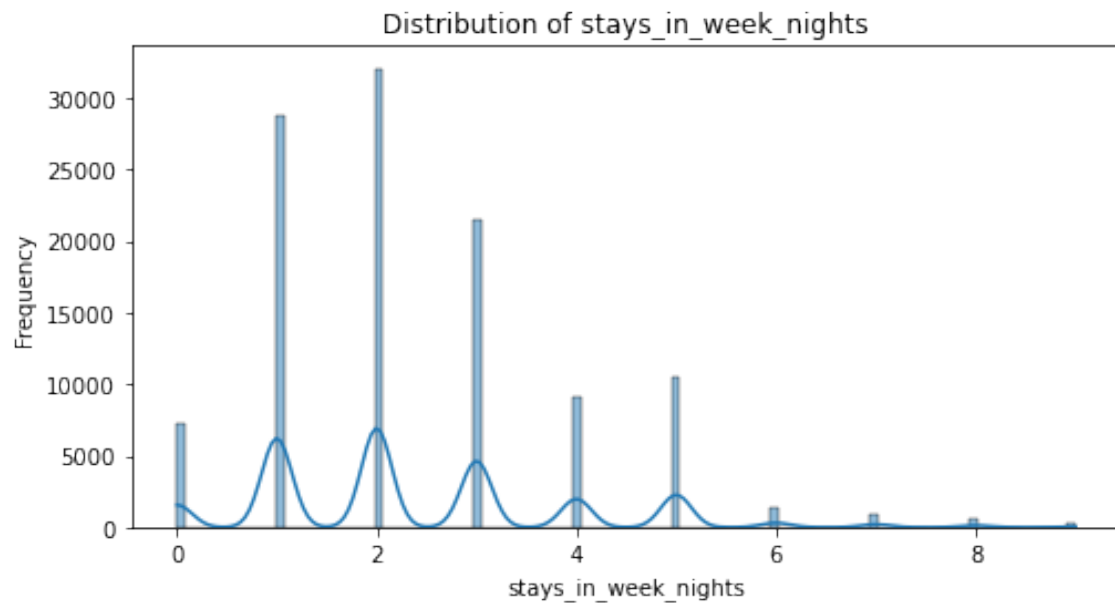


Figure 12: Distribution of stay in week night

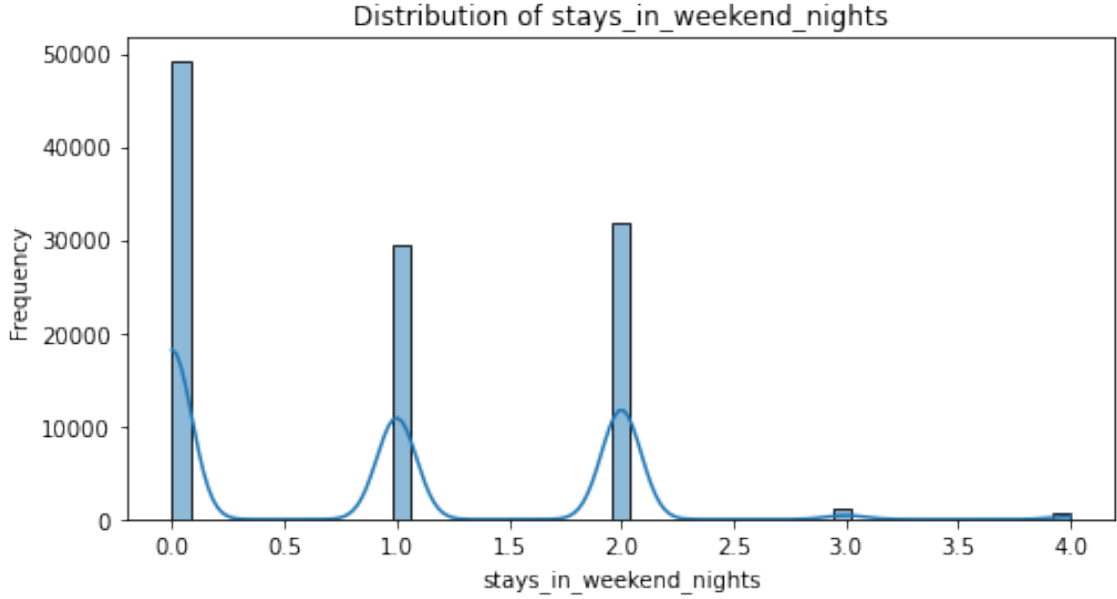


Figure 13: Distribution of stay in weekend night

- **total_nights and log_total_nights:** The total number of nights, combining weekend and weekday stays, showed a multimodal distribution. A logarithmic transformation was chosen over a squared one to better suit the count data and multiple peaks. This approach helps the model handle the non-linear relationship between stay length and cancellation probability without overstressing longer, less frequent stays.

After generating new variables through interaction terms and initial non-linear transformations, further non-linear transformations were applied to enhance model precision:

- **Interaction Term lead_time_x_market_segment_Corporate:** Employed Box-Cox to correct right-skewness, aligning this variable more closely with normal distribution assumptions.
- **Variable special_requests_x_customer_type_Transient:** Applied a log transformation to normalize its right-skewed distribution, directly aiming to evaluate how special requests among transient customers impact cancellation rates.
- **Variable previous_cancellations_x_booked_by_agent:** We applied a square root transformation to mitigate right-skewness and preserve the structural integrity of the data.

4.3 Logistic Regression with Interaction Terms and Non-linear Transformations

The logistic regression model was then refitted with these interaction terms and non-linear transformations. The performance metrics are presented in Table 3.

Table 3: Performance metrics for the Logistic Regression with Interaction Terms and Non-linear Transformations

Model	Accuracy	Precision	Recall	F1 Score
LR w/ Int. & NLT	0.8157	0.8418	0.6300	0.7207

The incorporation of interaction terms and non-linear transformations in the logistic regression model led to a notable improvement in its predictive performance. The model's accuracy increased to 81.57%, and the precision reached 84.18%, indicating a higher proportion of correctly predicted cancellations and a lower false-positive rate. The F1 score, a balanced measure of precision and recall, also improved to 72.07%, confirming the overall enhancement in the model's performance. These

results highlight the effectiveness of capturing complex relationships between the predictors and the cancellation probability through interaction terms and non-linear transformations, ultimately leading to more accurate predictions.

5 Dimensionality Reduction and Regularization

To improve the logistic regression model, we explored dimensionality reduction using L1 (Lasso) and L2 (Ridge) regularization. These techniques prevent overfitting, reduce model complexity, and enhance generalization by adding a penalty term to the loss function during training. L1 regularization encourages sparsity and performs feature selection, while L2 regularization shrinks coefficients towards zero and handles multicollinearity.

We applied L1 and L2 regularization to the logistic regression model with interaction terms and non-linear transformations, tuning the regularization parameters using cross-validation. The performance metrics for the regularized models are presented in Table 4.

Regularization	Accuracy	Precision	Recall	F1 Score
LR w/ Int. & NLT & L1	0.8264	0.8499	0.6511	0.7373
LR w/ Int. & NLT & L2	0.8261	0.8449	0.6553	0.7381

Both L1 and L2 regularization improved the model’s performance compared to the non-regularized version. The L1-regularized model achieved an accuracy of 82.64%, precision of 84.99%, recall of 65.11%, and an F1 score of 73.73%, while the L2-regularized model obtained similar results. These findings suggest that regularization effectively reduced overfitting and improved the model’s generalization ability. L1 regularization also provided a more interpretable model by selecting the most relevant features.

6 Random Forest Classifier

We trained a Random Forest classifier as an alternative to logistic regression. Random Forests combine multiple decision trees to handle high-dimensional data and capture complex relationships.

A grid search was used to find the best hyperparameters: `n_estimators=500`, `logisticregression_C=0.01`, and `min_samples_split=2`. The performance metrics of the tuned Random Forest model are shown in Table 5.

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.8623	0.8611	0.7570	0.8057

The Random Forest outperformed the logistic regression models, with an accuracy of 86.23%, precision of 86.11%, recall of 75.70%, and an F1 score of 80.57%.

A feature importance analysis was conducted to identify the most influential features, as shown in Figure 14. The top 5 features were: `lead_time`, `avg_daily_rate`, `deposit_type_No_Deposit`, `deposit_type_Non_Refund`, and `arrival_date_day_of_month`.

The Random Forest classifier proved to be a powerful alternative to logistic regression, demonstrating superior performance in predicting hotel booking cancellations. Its ability to handle high-dimensional data and capture complex relationships, combined with the insights from feature importance analysis, makes it a valuable tool for understanding and predicting cancellation behavior.

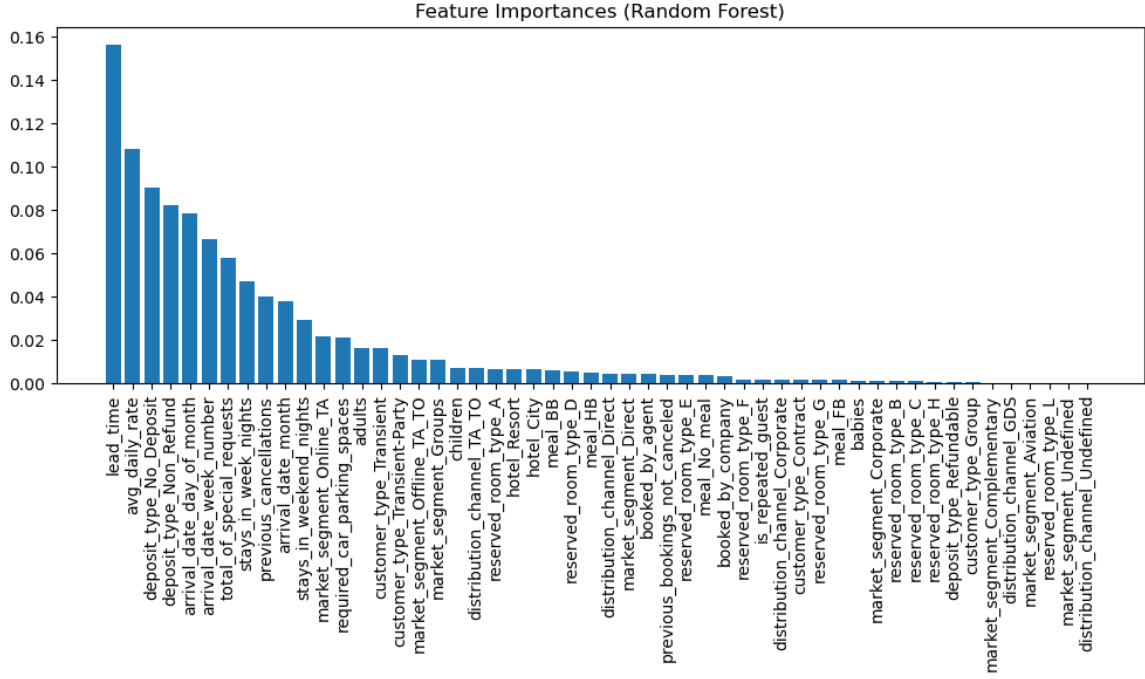


Figure 14: Feature importance for the Random Forest classifier

7 Model Comparison

Table 6 summarizes the performance of the models developed in this study. The baseline model, which predicts all bookings as not canceled, achieves an accuracy of 62.92%. The logistic regression models, with the incorporation of interaction terms, non-linear transformations, and regularization techniques, significantly outperform the baseline, with accuracies ranging from 80.35% to 82.64% and F1 scores from 70.57% to 73.81%.

However, the Random Forest classifier emerges as the best-performing model, with an accuracy of 86.23% and an F1 score of 80.57%. Its ability to capture complex patterns in high-dimensional data makes it the most effective model for predicting hotel booking cancellations.

Table 6: Performance comparison of different models

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.8035	0.8318	0.6128	0.7057
LR w/ Int. & NLT	0.8157	0.8418	0.6300	0.7207
LR w/ Int. & NLT &L1	0.8264	0.8499	0.6511	0.7373
LR w/ Int. & NLT &L2	0.8261	0.8449	0.6553	0.7381
Random Forest	0.8623	0.8611	0.7570	0.8057

In conclusion, while the logistic regression models provide valuable insights, the Random Forest classifier proves to be the most effective for predicting hotel booking cancellations, offering the highest accuracy, precision, and recall.

8 Executive Summary

In the competitive world of hospitality, understanding and predicting hotel booking cancellations is crucial for optimizing revenue and enhancing guest satisfaction. Our study dives deep into hotel booking data, uncovering key factors that influence cancellations and providing actionable insights for hoteliers. From our exploration into our dataset, we have several recommendations for hotel executives that could reduce the number of cancellations. Hotels should look to offer reservations that

must be booked at least a week in advance, including required deposits and reserved parking for guests. Encouraging employees to accommodate special requests and catering to guests with cars could also be beneficial. Marketing strategies should favor individual guests throughout the year, emphasizing the personal touch and convenience of direct bookings. By investing in direct communication channels rather than intermediary agents, hotels could enhance guest loyalty and reduce cancellation rates. Moreover, our findings indicate that cancellation patterns vary across different booking platforms and hotel types, necessitating customized approaches. For example, city hotels and resort properties show different cancellation trends, requiring hoteliers to develop tailored marketing and operational tactics. This study also suggests revisiting the practice of non-refundable deposits to align with modern guest expectations and foster a more guest-centric approach. By adopting these strategies, hoteliers can proactively address the issue of cancellations, refine revenue management practices, and offer personalized experiences that meet guest preferences. This data-driven approach positions hotels to make informed decisions, enhance profitability, and stay ahead in a dynamically changing market.